Meaning and Interpretation: An analysis of two theoretical perspectives in product design

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Abstract

Experiences gain increasing attention in design research, but theories are inconsistent, and the foundations of different methods are rarely discussed. Design research has *sui generis* no fixed epistemology, and is open to methodological eclecticism. However, as paradigms successively change from positivistic towards more comprehensive views, discussions on how underlying theoretical assumptions influence approaches are needed. This paper examines and compares axiological, epistemological and methodological aspects of two perspectives: Kansei engineering, and a postmodernist framework. Both frameworks aim to match designers' conceptions of product symbolism etc. to those of the user group, but they evolve from different theoretical starting points. Analyzing the contents of underlying theories and their influence on methods and expected results is an important part of theory development that may also support designers in making methodological choices.

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Introduction

In 1983, a secret meeting took place at the Toyota head office in Nagoya, Japan. The CEO, Eiji Toyoda asked his staff point blank: "Are we capable of building the best luxury sedan in the world?" The answer was "yes", and a plan for the 'Circle F' (for flagship) project was developed with the goal to compete directly against the BMW 7-Series, Jaguar sedans, and the Mercedes S-Class as a viable luxury car for the American market. One of the first activities of 'Circle F' was to figure out what luxury meant to Americans. Engineers and designers were send to live in the chic districts of Los Angeles to study in an almost anthropological way the behavior of rich Americans – not just their relationship to luxury cars but their lifestyle and consumer habits. Launched in 1989, the 'Lexus' soon became associated with quality, luxury and superior customer satisfaction and one of America's best-selling cars (Air France magazine, 2008, 99-100). By giving values such as 'beauty' and 'luxury' a higher priority than the traditional engineering requirements, the Toyota managers found a way to approach the consumer emotionally. What may have been revolutionary in the 80:ies is today widely accepted; emotions play a significant role in for consumer choices.

The scope of design has successively changed - from a focus on material aspects to a focus on the intangible, from functions to pleasure, from goods to services and values (Jones 1997, Pine & Gilmore 1998). Unfortunately, designers' conceptions of product symbolism, user experience etc. do not necessarily match those of the user group. Various methods have been proposed for treating these issues in a more 'objective' manner (see e.g. Desmet 2002, Green and Jordan 2002), indicating a desire to address complex relationships between people and things, rather than just giving form to dead matter. However, the theoretical foundations of these methods are rarely made explicit. Moreover, some imply philosophical anachronisms, such as naïve empiricism or materialism, assuming e.g. that the immediate sense experience is sufficient to provide knowledge, or that meaning could be 'designed into' a product.

Design has a rather small core of scientific "truths", and a rather wide area of "intruders"[1]. These intruders may be a threat to an established science. However, for design research they could provide an opportunity to learn from other disciplines and cumulate knowledge to advance

[1] We adapted the 'core' metaphor as theory of science approach for design from Imre Lakatos (1999). Very roughly: for Lakatos scientists involved in a research program will attempt to shield the theoretical core (fundus of theories) from intruding attempts from outside. While established fields such as physics and philosophy have big and protected cores, the core of design is small and can easily be intruded by other 'competing' theories, which is an advantage

design theory. Theoretical openness may contribute to both scientific and practical insights to the design field, which is traditionally addressed by experts possessing a lot of tacit knowledge but sometimes limited possibilities to articulate it. By taking the liberty to examine, criticize and modify existing methods, knowledge generation is stimulated. Experiences and methods from different areas such as engineering and social sciences can be adopted and adapted. However, the importance of understanding the underlying reasoning of different perspectives should not be underestimated as this affects what conclusions can be drawn from methods.

Out of many possibilities to assess or interpret users' attitudes and values towards products, this paper compares two perspectives in product design: Kansei engineering and a postmodernist framework[2]. Our intention is not to judge the value of these perspectives, but to make contrasts clear. The first section summarizes main characteristics of the two perspectives. The second section presents significant knowledge features of axiological (the normative sources of knowledge), epistemological (knowledge formation) and methodological (knowledge structuring) aspects. Conclusively, implications for the design field are discussed in section three.

Two perspectives

A main driver for inquiries in design is to 'translate' people's needs and wishes into desirable product attributes (Verganti, 2003). Scientific methods could seemingly provide control over this process by presenting alternatives to subjective judgments by designers, in the form of reliable data, derived from empirical studies.

One method that has gained wide attention in design is Kansei ('sense') Engineering (e.g. Nagamachi 1995, Schütte, 2005)[3]. In Kansei Engineering product properties are linked to sentiments by studying correlations[4] between products attributes and ratings of different

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(as far as these are reflected and absorbed), and a danger (if designers are not conscious about how these transform their discipline).

[2] Postmodernist thoughts can be traced in e.g. service design, design for all, social design, green design etc.

[3] Kansei typically relies on the semantic differential, proposed by Osgood, Suci & Tanembaum (1957) who developed metrics for the assessment of meanings of cultural objects explicated through a technique allowing participants to rate connotations of objects, persons or ideas with respect to a number of dimensions. It allows a quantification of to what extent an object was associated with an adjective at one pole or the other. By collecting ratings for several objects with respect to multiple dimensions it is possible to calculate how closely they are associated with each other. In design it has been used to e.g. evaluate the expression of products.

[4] Alternative types of data collection and analysis methods are sometimes used. As an example; Lee et al (2004) let users select images which were analyzed and translated into

’feelings‘(Nagamachi, 1995). This builds on a process where a number of competing products are collected and analyzed in terms of technical parameters (e.g. colours, dimensions of radii etc.). A large number of adjectives that may describe the product are also gathered. The adjectives are reduced by a small number of people by analyzing the products using semantic differentials with respect to all adjectives. By analyzing what adjectives co-vary, the number of adjectives can be reduced. A (larger) number of respondents are then asked to evaluate the products with respect to the remaining adjectives using semantic differentials. By analyzing correlations between technical properties and scores on the semantic scales, mathematic models can be developed that describe the semantic scores as functions of the technical parameters. These formulas are then applied in a prescriptive way. Given that we want to develop a product that gets a certain score on the scales we can derive what technical parameters to use to elicit a certain rating

In Kansei Engineering, meaning is more or less treated as an attribute of an object. Parallels can be drawn to industrial design methods from the 1960:ies. Both seem to rest on a positivist cannon assuming that objects exits in the real world, that they are observable by a neutral scientist, who can make claims that are verifiable through experiments and manipulate the objects with help of different techniques.

According to Swann (2002, 50), “Postmodernist philosophy - challenged this {engineering} dogma and urged a more tolerant and pluralistic approach to what might be good for the end-users. The social sciences brought forth a number of alternative ways to investigate and validate research and information, alternatives that have more affinity with design processes than the science/engineering model.”

The term ‘postmodern’, coined by Jean-Francois Lyotard, refers to a movement characterized by ‘incredulity towards metanarratives’ (Lyotard 1984, xxiii). Narrative elements disintegrate into ‘clouds’ of linguistic combinations and language games. The goal of discourses and actions is thus not to seek consensus and (hermeneutically) negotiate meaning (a metanarrative) but to conceive lack thereof as a productive mechanism for change. Postmodernism is beyond the belief of the existence of any ultimate scientific, philosophical, or religious truth[5]. Discursive interpretation becomes crucial; reality only comes into being through our reflections on what the world means to us individually and culturally. Postmodernism relies on concrete experience

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concepts which were then commented by users. Advanced expert systems have also been proposed which analyze data by the use of e.g. Fuzzy Logics.

[5] How far a truth claim is made in the assertion that there is no ultimate truth is not discussed here.

over abstract principles, knowing always that the outcome of one's own experience will necessarily be fallible and relative, rather than certain and universal. Because of contextual dependencies, the determination of the meaning of objects is ambiguous - especially when meanings are interpreted by different actors within a discourse.

Employing these thoughts for a design theory would mean to abandon a more or less tacitly accepted view on the determinability of users' feelings through a designer, and/or to acknowledge the inexplicability, contingency and polyvalence of meanings connected with products. It might also imply focusing on discourses (narratives) between designers and users rather than the attributes of the discussed object. As Richardsson (1993, 41) puts it: "…we need to find a new theory of form to describe the user-product nexus. It should be one that embraces the unpredictability of interpretations to be found in the use-place, one that takes a positive rather than negative attitude toward the flexibility of meaning in design's formal manifestations".

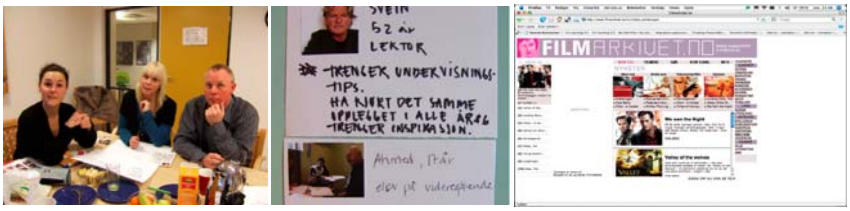


Figure 1. Example of the use of narratives and personas to organize the film archive at The Norwegian Film institute to make it available for the general public.

In design curricula the postmodernist framework can help students develop an understanding of different views, sensitizing them to users' experiences and values when designing a product. This was one goal of a learning lab, which took place at the Oslo School of Architecture and Design in spring term 2008 as part of a project called Ludinno. In this project, four Scandinavian universities, financed by the Nordic Innovation Centre, cooperated in testing innovation through users. The student groups were supposed to develop personas and narratives, employing the scheme presented in Table 1, use them throughout the design process and to communicate with the companies (see example in Fig. 1).

Context	**Qualitative examples**	**Quantitative examples**
Physical	Personal experience and conversations with relatives, interviews	Survey of older persons and health
Mental	Conversations, articles, stories, movies	
Intellectual	Literature, Novels, Movies	Reports on intellectual development
Social	Interviews, Literature	Survey of social situations, Statistics
Ethical	Own experience and attitudes	
Aesthetic	Exposure of persons in media,	

	Appearance of existing products	
Specific	Interaction with electronic devices, Advertisements	Technical reports, Benchmarking existing products

Table 1: Addressing interface design for older users.

Comparison of Kansei engineering and the postmodernist approach

One analytic way of describing a research program is to examine its axiological and epistemological aspects, to trace how these influences and contribute to justify methods, and to look at the implications these have.

With ***axiological aspects*** (Gk. *axios* - worth and *logos* - speech, word, reason, Merriam-Webster 2008) we mean underlying assumptions about the world within which the researcher builds her hypotheses and chooses certain methods to obtain insights, i.e. generates knowledge. ***Epistemology*** (Gk. *epistanai* - to understand, know, Merriam-Webster 2008) refers to the study or theory of knowledge, especially with reference to its limits and validity. ***Methodology*** (Gk. *meta*- after and *hodos* -a traveling, way - to pursuit, following after, Etymology Dictionary 2008) describes here the analysis of methods employed.

	Kansei	**Postmodernist framework**
Theory tradition	Natural sciences, engineering	Humanities, Social sciences
Source of the experience (gradually moving from extensive to intensive distinctiveness)	Embedded in the artefact	Embedded in culture
Relationship between Designer and User	Detachment	Involvement, Empathy
Axiology	Quasi-causal relation between object and meaning	Objects exist through interpretation (design as ”language”)
Epistemology	Knowledge generation through observation, empiric, nomothetic	Knowledge generation through interpretation, idiographic
Methodology	Instrumental research, cook bookish approaches giving advices on ‘best practices’	Textual analysis

Table 2: Comparison of the theoretical foundations of two approaches.

Kansei engineering ***axiology*** is characterized by the belief that objects and meaning stand in a quasi causal relationship (e.g. in Kumaou et al. 2005, Camurri et al. 2004). Kansei proponents seem not to consider that attributing meaning to products may be a dynamic process of interplay between the user’s understanding and the product. Moreover, Kansei accepts a more or less tacit view on the general interpretability of consumers’ preferences through a designer, who is to a certain degree detached from the users. This may be a heritage from the natural sciences where

methodological justification is achieved through the scientist's conduct of controlled experiments, identifying cause-and-effect relationships, while remaining strictly neutral and disconnected from personal views.

The theoretical reasoning of Kansei ***epistemology*** stands rather disproportional to its methods. As an example, Kansei Engineering would claim it possible to measure the user's feeling of "good taste" by letting him taste a new dish, assess his reaction and rationalize this with empiric means without providing explanations. In measuring peoples' believes Kansei seems to rely on a justification scheme which emphasizes data through sense experiences as the supreme source and criteria of knowledge. Kansei also reveals naïve empiricism[6] in form of treating words more or less like "objects…situated in a semantic field, which has both universal and individual components" (Douglas 1996). However, words such as 'sporty' 'ugly' etc. are neither existing entities, nor inherent in the object. Words imply cultural interpretations and symbolic practice, which have to be understood to connect them with objects (Keitsch 2007). For example, the word 'Venus' may stand as denotation of the planet and as a metaphor for 'beauty' while both words are not the Venus itself. To understand and use the word, one has to be part of a certain discourse including e.g. the Western idea to observe celestial objects and to name these after of gods from Greek mythology.[7]

Kansei ***methodology*** is partly focused on the search for correlations, which may determine the degree and direction of relationship between two or more variables. An advantage of the Kansei methodology is its capability to reduce information, focusing on important dimensions. While the information in a quantitative study will lose in richness, it enables studies across groups. Interviews etc. concerning metaphors and symbolism may support designerly understanding of how a specific user or a group of users attribute meanings and symbolism to something, but it is sometimes desirable to have more specific criteria and measures. Furthermore, evaluation of competing products and concepts may help the designer to identify features that at least

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[6] Naive empiricism claims that humans have to have experiences before they can say something about the world. However, each experience is based on a fundament of expectations and hypotheses, namely those that constitute our horizon and that make these experiences meaningful, see also: Popper (1998, 359).

[7] Wittgenstein (1967, § 116) sees this in a language context: "A sign lives in the language-game that is its original home." This seems also valid for non-linguistic signs see e.g. Moore (2003, 34): "When a particular version of the visual language is taken out of its context – the medium, the time, or the culture – it fails" (2003, 34). See also: Hiort af Ornäs, Persson, & Jordan (2007): "Meaning ascribed to a thing can hence neither be explained by the thing alone, nor by the product in combination with prior experiences…. Take the concept of 'utility' which seems to imply a negative valence when taken as an opposite of 'pleasurable', but is positive when contrasting 'useless'. Similarly 'design' was used to describe means for making something aesthetically pleasing but in other cases as something shallow, an attempt of value-creation through something superfluous."

correlate with certain attributed meanings, allowing some tentative conclusions. However, correlation does not reveal whether or not a relationship is a cause-and-effect one, and unpredictability has to be taken into account.

A challenge for Kansei Engineering is that the validity is highly depending of sampling; of products, participants, adjectives etc. which need to reflect the reality one wants to make claims about. Furthermore, the successful analytical identification of correlations between technical parameters and adjectives is by no means a guarantee for successful synthesis. The mathematic analysis of the collected data may provide information on the 'whats' and 'how muches' of meaning. What is also needed, however, is how meaning is interpreted and connoted by different users, and that is where the postmodernist framework becomes valuable.

The ***axiological*** underpinnings of the postmodernist approach can for our purposes be summarized as acknowledging the importance of discourse and text interpretations, and awareness of creative interaction possibilities between different participants.

One main ***epistemological aspect*** of this perspective is the interpretation of a text. Text means here not only written texts, but also artefacts, relationships etc. (Richardson 1993, 36). Language is seen as “a form of social practice” (Fairclough 1989, 20) and knowledge generation has the goal “to unpack the ideological underpinnings of discourse that have become so naturalized over time that we begin to treat them as common, acceptable and natural features of discourse“ (ibid). For design research this means methodologically e.g. examining the social practices of a certain culture where a discourse takes place by interacting with people in their everyday settings, and then potentially proceed to identify specific user groups and artefacts for further studies.

Methodologically, the postmodernist framework can be related to hyper-linguistic or supra-linguistic analyses, where researchers reflect on the larger discourse context or on meaning that lies beyond the language structure. This includes a systematic consideration of social, political, aesthetic and economic frameworks. If the textual analyses are clear and consistent, the postmodernist framework presents a qualitative view of a complex situation and, especially important, of users’ values, feelings and probably their hidden agenda, which can be relevant for design processes. Further, many postmodernist authors are not content to describe models and orders for existing phenomena but use an almost poetic language to approach those that can enlighten the discourse about both the phenomena and the methods. On example is Deleuze’s use of the ‘rhizome’ metaphor (an underground plant stem whose growth is aimless and disordered). The ‘rhizome’ works as a anti-metaphor to the “arboreal” (hierarchical, orderly, and linear) character of conventional philosophies and as a methodological mean to argue for

multiple, non-hierarchical data representation and interpretation in theory and research: "We're tired of trees… unlike trees or their roots the rhizome connects any point to any other point, and its traits are not necessarily linked to traits of the same nature; it brings into play very different regimes of signs, and even nonsign states. The rhizome is reducible neither to the One nor the multiple... It is composed not of units but of dimensions, or rather directions in motion." (1987, 15, 21)

One pitfall of postmodernist analysis is that general characteristics of products are difficult to comprehend if the analysis is subjectivistic and not able to combine individual and shared impressions of phenomena. Another drawback is that postmodernists' axiology shrinks experiences to questions about interests and interpretations of power. Finally, any approach criticizing theories that determine 'identity' by using 'definitions' or 'models' should be able to give preponderance to the 'object' before its 'connotation'. Postmodernism remains in this sense (through its discourse methodology) bound to the rationality it is criticizing (See also Vihma 2008).

Implications

"…all-explanatory theories have an irresistible effect on the weak mind."

Karl Popper

Assuming meaning as an attribute of an object leads to seeing it as a metric. Meaning is then measured and averaged. The relation to users and study participants is here one of detachment. The downside is that the focus on precising measurements may take over in favor of the concern for understanding what different measurements stand for. Assuming interpretation of products based on their highly individual experience means more or less abandoning the idea of patterns, implying a methodology where focus is on engagement with users and their experiences, interpreting and developing an idea about what is important but not generalizing any findings. The downside of this perspective is that the information is imprecise, and not easily communicated or put on trial. The challenge for the Postmodernist framework is to realize that even the most individual experience may under certain circumstances give access to phenomena of a truly shared character such as emotional involvement in dealing with objects, phenomena which are, although in several senses not at all new, certainly now growing in importance and quantity across the globe.

The two perspectives probably have merits, and weaknesses relating to practice, but distinctions go deeper than that and touch the underlying assumptions and world-views.

From our point of view design has a normative raison d'être, and design research is advantageous whenever it enables designers to make well-founded decisions. A special characteristic of design research is that discussions draw on knowledge from many fields. It is not possible to find one coherent paradigm or an everlasting theory. This scientific position comes with challenges, especially if researchers are not aware of what kind of theories they adopt but it provides also advantages in many cases when comprehensive design views are required.

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